

Part A – Fundamentals

1. What is Data Analysis?

Data Analysis is the process of collecting, cleaning, transforming, and interpreting data to discover useful information and support decision-making.

The main goal of data analysis is to identify patterns, trends, and relationships within data.

Steps in Data Analysis:

1. Data Collection
2. Data Cleaning
3. Data Exploration
4. Data Visualization
5. Interpretation of Results
6. Decision Making

Importance of Data Analysis:

- Improves business decisions
- Identifies customer behavior
- Predicts future trends
- Supports machine learning applications

Example:

A company can analyze customer purchase behavior to understand which customers are likely to stop purchasing products.

2. Data Science Project Life Cycle

A Data Science Project follows a structured process from problem definition to deployment.

Step 1: Problem Understanding

Understand the business problem and project objectives.

Step 2: Data Collection

Gather data from various sources such as:

- CSV files
- JSON files
- SQL Databases
- APIs

Step 3: Data Cleaning

Handle:

- Missing values
- Duplicate records
- Incorrect data types
- Outliers

Step 4: Exploratory Data Analysis (EDA)

Analyze and visualize data to understand patterns and relationships.

Step 5: Feature Engineering

Create new useful features from existing data.

Step 6: Data Preprocessing

Prepare data for machine learning:

- Encoding
- Scaling
- Normalization

Step 7: Model Building

Train machine learning models.

Step 8: Model Evaluation

Evaluate model performance using metrics.

Step 9: Deployment

Deploy the model into a real-world application.

Step 10: Monitoring and Maintenance

Continuously monitor and improve the model.

3. Machine Learning Problem Statement

Customer Churn Prediction

Problem Statement

The company wants to predict whether a customer will churn based on purchase behavior and customer information.

The dataset contains:

- Age
- Gender
- Income
- Purchases
- Purchase Frequency
- Last Purchase Days
- Membership Status
- City

Objective

Develop a machine learning model that predicts whether a customer is likely to leave the company.

Input Features

- Age
- Gender
- Income
- Purchases
- Purchase Frequency
- Last Purchase Days

- Membership Status

Target Variable

Churn

Possible values:

- Yes
- No

Benefits

- Improve customer retention
 - Reduce revenue loss
 - Increase customer satisfaction
 - Support business decision-making
-

4. What are Tensors?

A Tensor is a mathematical object used to represent data in multiple dimensions.

Tensors are the fundamental data structure used in:

- Artificial Intelligence
- Machine Learning
- Deep Learning

A tensor can be considered a generalization of:

- Scalars
 - Vectors
 - Matrices
-

Types of Tensors

0-D Tensor (Scalar)

A single numerical value.

Example:

5

Shape:

0

1-D Tensor (Vector)

A collection of values arranged in a single dimension.

Example:

[10, 20, 30]

Shape:

(3,)

2-D Tensor (Matrix)

Rows and columns of values.

Example:

```
[  
  [1, 2],  
  [3, 4]  
]
```

Shape:

(2,2)

3-D Tensor

A collection of matrices.

Example:

```
[  
  [  
    [1,2],
```

```
[3,4]
],
[
[5,6],
[7,8]
]
]
```

Shape:

```
(2,2,2)
```

In-Depth Explanation of Tensors with NumPy Examples

NumPy arrays are commonly used to represent tensors.

Example 1: Scalar Tensor

```
import numpy as np
```

```
scalar = np.array(10)
```

```
print(scalar)
```

Output:

```
10
```

Example 2: Vector Tensor

```
vector = np.array([10,20,30])
```

```
print(vector)
```

Output:

```
[10 20 30]
```

Example 3: Matrix Tensor

```
matrix = np.array([
    [1,2,3],
    [4,5,6]
```

```
])
```

```
print(matrix)
```

Output:

```
[[1 2 3]
 [4 5 6]]
```

Example 4: 3D Tensor

```
tensor3d = np.array([
    [[1,2],[3,4]],
    [[5,6],[7,8]]
])
```

```
print(tensor3d)
```

Output:

```
[[[1 2]
 [3 4]]

 [[5 6]
 [7 8]]]
```

Why Tensors are Important in Machine Learning?

Tensors help store and process large amounts of multidimensional data efficiently.

Applications:

- Image Processing
- Natural Language Processing
- Recommendation Systems
- Deep Learning Models
- Neural Networks

Examples:

- Grayscale image → 2D Tensor

- Color image → 3D Tensor
- Video → 4D Tensor

Tensors are the foundation of modern AI frameworks such as TensorFlow and PyTorch.

Conclusion

Data Analysis helps organizations understand data and make informed decisions. A Data Science project follows a structured workflow involving data collection, cleaning, analysis, modeling, and deployment. Customer churn prediction is an important machine learning problem that helps businesses retain customers. Tensors provide a powerful way to represent multidimensional data and are widely used in machine learning and deep learning applications.